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| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 07/25/2018 |
| **PI Name:** | Hyers, Robert Ph.D. | | |
| **Project Title:** | Modeling and Simulation of Electrostatically Levitated Multiphase Liquid Drops | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Materials science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | rwhyers@wpi.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 508-831-5759 |
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| **City:** | Worcester | **State:** | MA |
| **Zip Code:** | 01609-2280 | **Congressional District:** | 2 |
| **Comments:** | Formerly with University of Massachusetts at Amherst. | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2012 Japanese Space Agency (JAXA) AO for International Research Participation on ISS Kibo Module |
| **Start Date:** | 04/08/2014 | **End Date:** | 04/07/2019 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA MSFC |
| **Contact Monitor:** | Craven, Paul | **Contact Phone:** | 256-544-7649 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | July 2018 report: Prof. Jonghyun Lee moved to Iowa State University to accept a tenure-track position in September 2017. Per discussions with NASA, efforts are underway to continue his participation through a subaward. | | |
| **COI Name (Institution):** | Lee, Jonghyun Ph.D. ( Iowa State University (formerly University of Massachusetts) )<br>Watanabe, Masahito Ph.D. ( JAXA PI: Gakushuin University ) | | |
| **Grant/Contract No.:** | NNX14AH85G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Funding is for Dr. Hyers's role as U.S. Principal Investigator (PI) on Professor Masahito Watanabe's (Department of Physics, Gakushuin University, Tokyo, Japan) proposal, “Interfacial Phenomena and Thermal Properties of High-Temperature Melts,” funded by the Japanese Aerospace Exploration Agency (JAXA). Prof. Masahito Watanabe's project: During steel making process, such as continuous casting, the impurity in the cast steel is influenced by the interplay between the molten steel and molten slags. Understanding the interfacial phenomena is necessary to produce high-purity steels. An accurate measurement of interfacial tension between molten steel and molten slag is one essential step to understand the interfacial phenomena. Well-controlled oscillation of a sample with a concentric core (molten steel) – shell (molten slag) structure is required to measure the interfacial tension. Due to the effect of gravity, this concentric arrangement cannot be obtained using ground-based facilities. An Electrostatic Levitation Furnace (ELF) in the KIBO deck in the International Space Station (ISS) will be utilized for the measurement of interfacial tension at the interface between molten steel and molten slag. For the success of the space mission, it is of critical importance to predict the interfacial tension and the influence of main parameters, such as the core-shell ratio, thermophysical properties of molten slags, magnitude of initial deformation, magnitude of electrostatic forces, and off or inphase oscillation, on the predicted interfacial tension. With such information, the space experiment can be designed efficiently and effectively.<br>Dr. Hyers's (University of Massachusetts) scope of work as part of Dr. Watanabe’s proposal is summarized below:<br><br>1. Simulation of the flow in electrostatically levitated core-shell droplets, including dynamics such as oscillations and viscous effects on the dynamics.<br><br>2. Simulation of the effect of changes in the dimensions of the core and shell on the surface oscillations.<br><br>3. Simulation of the time dependence of the changes in surface oscillations due to reactions between the oxide and steel phases.<br><br>4. Design, direct, and analyze experiments to be conducted in the Marshall Space Flight Center (MSFC) Electrostatic Levitation Facility.<br><br>The Research is to use data from an Electrostatic Levitator (ESL) facility supplied by the Japanese to be used on the International Space Station, and also a ground based ESL facility located at MSFC. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The results of the project are expected to be utilized for more efficient production of high-purity steels, which are the main metals used on Earth. |
| **Task Progress:** | • Face-to-face meeting with JAXA PI Prof. Watanabe to discuss latest experimental results from parabolic flight campaign.<br>• On-site meeting with NASA personnel at MSFC, December 2017.<br><br>• A single phase droplet oscillation was successfully modeled with the volume of fluid method using ANSYS Fluent and it is being extended to simulate multiphase droplets.<br><br>• Presented results at the 146th Minerals, Metals & Materials Society (TMS) Annual Meeting in Phoenix, AZ, February, 2017. |
| **Bibliography Type:** | Description: (Last Updated: 03/30/2026) |
| **Books/Book Chapters** | Allamore A, Lambotte G, Lee J. (editors) "Materials Processing Fundamentals 2017." Ed. A. Allanore, G. Lambotte, J. Lee. Cham: Springer, 2017. The Minerals, Metals & Materials Society. https://doi.org/10.1007/978-3-319-51580-9 , Feb-2017 |
| **Papers from Meeting Proceedings** | Sumaria K, Hyers TW, Lee J. "Oscillation of a zirconium droplet – experiment and simulations." 146th TMS Annual Meeting, San Diego, CA, February, 2017.<br>In: Materials Processing Fundamentals 2017. Allanore A., Lambotte G., Lee J., eds. (The Minerals, Metals & Materials Series) Cham: Springer, 2017. pp.65-72. https://doi.org/10.1007/978-3-319-51580-9_7 , Feb-2017 |